

Benchmarking Factual Accuracy of Large Language Models Across Scientific Domains Abstract

Large Language Models (LLMs) demonstrate remarkable capacities across natural language processing tasks, yet their susceptibility to factual hallucinations poses significant risks in domain-specific academic and scientific applications. This paper presents a synthesis and benchmark analysis evaluating the factual correctness, domain resilience, and hallucination modalities of state-of-the-art LLMs across core scientific disciplines: Medicine, Chemistry, Physics, and Computer Science. Utilizing standard benchmarks alongside fine-grained error tracking datasets (e.g., FELM, RealFactBench, TruthfulQA), we analyze how model scale, retrieval-augmentation mechanisms, and domain complexity influence factual accuracy. Our analysis highlights that while instruction-tuned and retrieval-augmented systems mitigate general surface-level errors, dense scientific domains featuring complex quantitative reasoning and specialized terminologies exhibit persistent error rates. We conclude by identifying key technical gaps in automated fact-checking, knowledge drift, and multi-step domain verification, offering strategic directions for future benchmark engineering.

Keywords: Large Language Models, Factual Accuracy, Hallucination, Scientific Information Extraction, Benchmarking, Information Retrieval.

Introduction

The integration of Large Language Models (LLMs) into scientific workflows—ranging from literature discovery and automated hypothesis generation to peer review assistance—demands an uncompromised level of factual fidelity. While LLMs exhibit fluent generation and human-like contextual reasoning, their underlying autoregressive token prediction mechanism is fundamentally vulnerable to generating content that is syntactically coherent yet factually false (Lin et al., 2022). In scientific contexts, such errors can result in severe consequences, including misinformed clinical decisions, flawed chemical synthesis protocols, or erroneous theoretical proofs.

Evaluating factual accuracy across specialized domains requires moving beyond standard open-domain trivia metrics. Scientific knowledge is dense, highly structured, rapidly evolving, and heavily dependent on multi-step reasoning. Consequently, benchmarking LLMs on scientific facts presents unique evaluation challenges: automated string-matching metrics fail to capture nuanced correctness, whereas expert human evaluations are costly and difficult to scale.

This study systematically investigates the landscape of factual accuracy benchmarking for LLMs across scientific disciplines. We examine evaluation methodologies, quantify domain specific accuracy variations, explore error mitigation techniques such as Retrieval-Augmented Generation (RAG), and outline open challenges in constructing reliable factuality evaluators.

Background and Literature Review

Factuality vs. Fluency in Generative LLMs

Factuality evaluation focuses on whether a model's generated output aligns with verified world knowledge and domain-specific ground truth (Chen et al., 2023). Early benchmark efforts, such as TruthfulQA (Lin et al., 2022), established that increasing model size alone does not

automatically eliminate factual inaccuracies; in fact, larger models can mimic widespread human misconceptions more persuasively—a phenomenon known as imitative falsehood.

Scientific Knowledge Representation and Hallucinations

Scientific domain errors typically manifest in distinct forms:

- Entity/Concept Misattribution: Conflating properties of distinct scientific entities (e.g., swapping binding affinities between isomeric chemical compounds).
- Quantitative Inaccuracy: Generating incorrect constants, measurements, or numerical outcomes during scientific problem solving.
- Reasoning/Logic Faults: Providing accurate raw premise facts but deriving scientifically invalid conclusions.

- Fabricated Citations: Generating plausible-sounding authors, titles, and DOIs that do not correspond to existing scientific literature.

Evolution of Fact-Checking Benchmarks

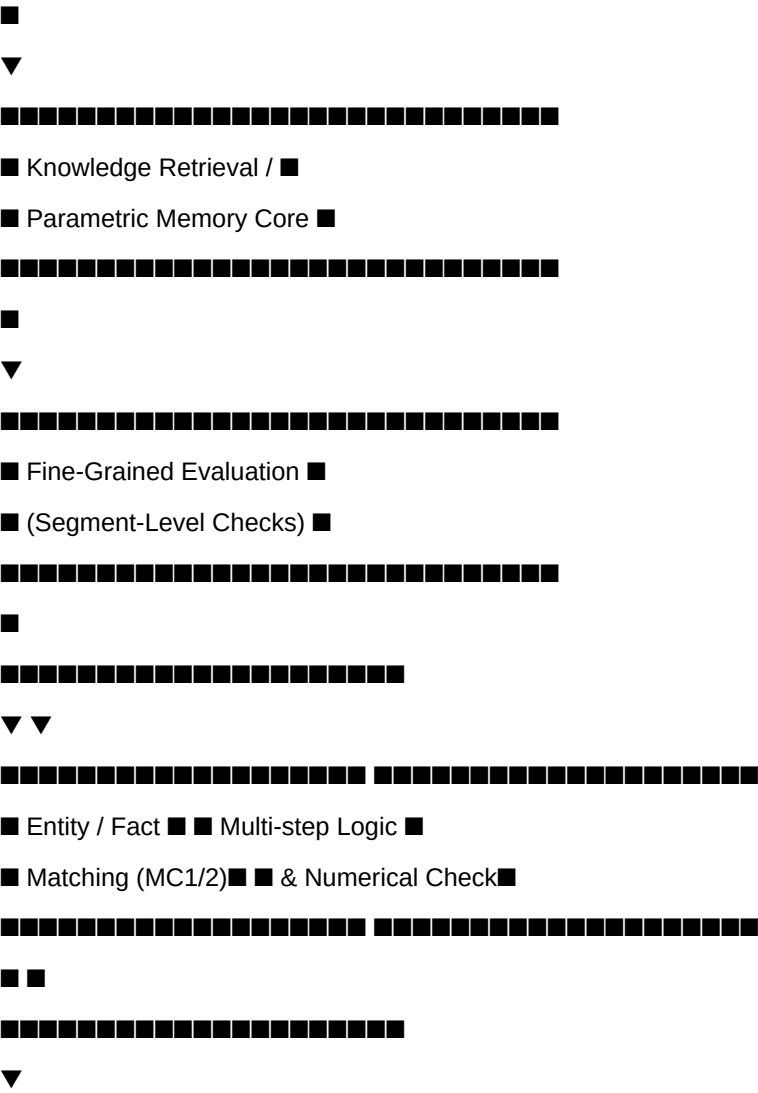
The trajectory of factuality benchmarks has evolved from static, closed-world question answering formats to dynamic, fine-grained, and multimodal frameworks:

- TruthfulQA (Lin et al., 2022): Tested model truthfulness against common human misconceptions across health, science, and general knowledge.
- FELM (Chen et al., 2023): Introduced segment-level annotations across diverse domains, including math and reasoning, isolating specific factual errors rather than relying solely on overall document-level scores.
- RealFactBench (Yang et al., 2025): Expanded benchmarking to real-world fact checking tasks, combining multimodal content, dynamic external knowledge, and retrieval capabilities across political, medical, and scientific domains.

Main Technical Discussion & Comparative Analysis

Evaluating LLM factual accuracy across scientific domains requires tailored datasets that target domain-specific representations.

[Input Scientific Query]



[Factuality Score & Error Map]

Domain-Specific Accuracy Characteristics

- **Medicine & Life Sciences:** Highly reliant on strict terminology and validated clinical protocols. Evaluators frequently leverage datasets like PubMedQA and BioASQ. RAG mechanisms significantly lower hallucinations by grounding generations in verified biomedical corpora.
- **Chemistry & Materials Science:** Requires structural accuracy (e.g., SMILES representation) and accurate property predictions. Models often suffer from structural hallucinations when generating molecular notation or reaction mechanisms.
- **Physics & Mathematics:** Requires rigid adherence to quantitative logic and symbolic computation. Models often fail not on factual knowledge retrieval, but on maintaining numeric and logical consistency across extended multi-step reasoning chains.
- **Computer Science & Software Engineering:** Evaluated via functional correctness (e.g., execution of code snippets) and execution trace verification, offering programmatic ground-truth feedback.

Key Benchmark Frameworks Matrix

Current Developments & Mitigation Strategies

To combat factual divergence in scientific generation, researchers employ several strategies:

1. **Retrieval-Augmented Generation (RAG):** Integrating real-time retrieval over scholarly databases (e.g., arXiv, PubMed) reduces reliance on static parametric memory. However, studies (Chen et al., 2023; Yang et al., 2025) show that while RAG improves empirical factual recall, models remain susceptible to incorrect context synthesis or misinterpreting retrieved scientific tables and figures.
2. **LLM-as-a-Judge Factuality Evaluators:** Utilizing automated LLM judges with targeted rubrics enables scalable fact-checking. Nevertheless, evaluators must be systematically calibrated to prevent judge bias and prevent the automated acceptance of plausible-sounding scientific errors.
3. **Chain-of-Thought (CoT) and Self-Consistency:** Prompting models to decompose complex scientific queries into explicit intermediate reasoning steps improves logical oversight and reduces arithmetic/conceptual failure rates.

Research Challenges, Gaps, and Future Directions

Persistent Challenges & Limitations

- **Temporal Knowledge Drift:** Scientific knowledge evolves rapidly. Parametric memory becomes outdated, and static benchmarks quickly suffer from temporal decay.
- **Data Contamination:** Benchmark items frequently leak into massive pre-training web scrapes, leading to artificially inflated accuracy scores that fail to reflect true reasoning capability.
- **Multimodal Scientific Reasoning:** Real-world scientific information is densely packed into figures, charts, and structural diagrams. Current text-only factuality frameworks fail to assess multimodal synthesis accurately (Yang et al., 2025).

Future Directions

- Development of dynamic, continuously updated benchmark testbeds that query newly published research papers to prevent contamination.
- Implementation of fine-grained, segment-level automated factuality checks that evaluate logical consistency alongside numerical/formulaic correctness.
- Standardizing multimodal factual evaluation to simultaneously test diagram comprehension, symbolic reasoning, and scientific context retrieval.

Conclusion

Measuring and ensuring the factual accuracy of Large Language Models across scientific domains is a foundational prerequisite for their adoption in research environments. While instruction tuning, specialized pre-training, and retrieval grounding have advanced performance on static QA benchmarks, evaluating complex scientific reasoning remains a critical challenge. Future progress requires fine-grained evaluation paradigms, robust multimodal fact-checking frameworks, and continuous benchmark updating to ensure that language models reliably support scientific discovery.

References

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